
EDUCATION

University of Washington (Seattle, WA)

Ph.D. in Computer Science & Engineering

Jun 2030

GPA: 3.90 / 4.0

Carnegie Mellon University (Pittsburgh, PA)

Master of Human-Computer Interaction in School of Computer Science

May 2025

GPA: 3.96 / 4.0

Howard University (Washington, DC)

B.S. in Computer Science in College of Engineering and Computer Science

Dec 2023

GPA: 3.79 / 4.0

PUBLICATIONS & PRESENTATIONS

1. **Layne, E. & Cakmak, M.** (2026). "Humanoid Robots in the Wild: A Digital Ethnography via Social Media Videos." *IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. (decision pending)
2. **Layne, E. & Cakmak, M.** (2026). "Humanoid Robots in the Wild: A Digital Ethnography of Online Videos." Special Session on "HRI and Consumer Robots: Building Robots that are Useful and Acceptable" (Sponsored by Meta), ACM/IEEE International Conference on Human-Robot Interaction 2026.
3. Radka, B., **Layne E., & Cakmak, M.** (2025). "Attitudes towards Humanoid Robots for In-Home Assistance." *IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*.
4. **Layne E.** and Mostow, J. "Replicating Bugs Faster". *Working Papers Journal 9*, pp. 155-156 (2021).

RESEARCH EXPERIENCE

University of Washington

Human-Centered Robotics Lab (Advisor: Maya Cakmak)

Sep 2025 – Present

- Conducted digital ethnography of 77 YouTube videos of Unitree G1 humanoid robots in naturalistic settings; identified widespread personalization (80.5%), physical contact in nearly 1 in 3 interactions, and intimate proximity as the dominant interaction distance, informing design recommendations for consumer humanoids
- Researching privacy concerns for consumer robots in home environments through literature review, examining paradigm shifts in privacy frameworks from 2015 to present

Massachusetts Institute of Technology Lincoln Laboratory

Group 42 - Transportation Safety and Resilience - *Full Secret Clearance*

Jun 2025 – Present

- Designing and implementing interactive data visualization dashboards that transform intricate simulation datasets into user-friendly graphs and visual displays, facilitating enhanced analysis of agent-based model simulations.

National Aeronautical Space Administration

Ames Research Center, Earth-Independent Operations

Jan 2025 – July 2025

- Leading technical design efforts to develop autonomous problem-solving processes for Mars-bound crew members, addressing communication delays while balancing vehicle constraints and UI feasibility requirements.
- Designed and evaluated dashboard-based solutions for deep-space missions, integrating insights from analogous domains to optimize real-time decision-making and problem-solving capabilities.
- Developed multiple Swift-based prototypes utilizing Anthropic API integration to test interface robustness and enhance crew situational awareness during mission-critical scenarios.

Carnegie Mellon University

eHeart Lab (PI: Geoff Kaufman)

Aug 2024 – Dec 2024

- Conducted user research using HCI methods to support ESL learners and facilitators in a VR-based Oculus environment, leveraging co-design and past research to guide development.
- Designed research plans and synthesized insights with Miro, integrating findings from prior studies to inform innovative approaches for immersive language learning.

RoboTutor (Advisor: Jack Mostow)

Jun 2021 – Aug 2021

- Developed a bug replication process with screen recording and JSON-based error tracking

INTERNSHIP EXPERIENCE

Apple Inc

Watch Complications (*Software Engineering Intern*)

May 2024 – Aug 2024

- Implemented a watch face using Objective-C with the MVVM architecture, utilizing Swift as the data source
- Added regional time display, gradient backgrounds, color themes, and numeral styles for enhanced personalization

Health Sensing Experience (*Software Engineering Intern*)

May 2023 – Aug 2023

- Designed and implemented multiple UI iterations for cycle deviations, using Swift and SwiftUI to enhance usability and support informed patient-doctor discussions
- Analyzed and worked with pre-existing cycle tracking data to determine the most relevant and non-alarming information for display, ensuring a balanced, user-friendly experience

Watch Frameworks (*Software Engineering Intern*)

May 2022 – Aug 2022

- Partnered with the Human Interface team to design an accessible watch experience for color-blind users, ensuring inclusivity and usability
- Developed multiple SwiftUI iterations and independently drafted an API proposal to enhance framework functionality and design flexibility

National Aeronautical Space Administration

Jet Propulsion Laboratory (*Reference Mission Systems Engineering Intern*)

Jun 2021 – Aug 2021

- Created and documented a step-by-step guide using Jupyter Notebooks to help employees without Linux or terminal experience navigate and resolve merge conflicts effectively
- Designed and implemented a structured system to streamline engineer update meetings, improving communication efficiency and reducing time spent on status reports

TEACHING & MENTORSHIP

Google Developer Student Club, President

Jul 2023 – Aug 2023

- Led technical programming and community-building efforts that connected underrepresented students to software engineering resources and industry networks

Paul G. Allen School of CSE, Research Mentor

Jan 2026 – Present

- Mentoring 2 undergraduates on privacy precautions for humanoids in domestic environments, study currently in progress

AWARDS & FELLOWSHIPS

Graduate

Black in Robotics Student Conference Travel Grant	2026
Herbold Fellowship	2025
University of Washington College of Engineering Dean's Fellowship	2025
National GEM Consortium Fellowship (MIT Lincoln Lab)	2025
National GEM Consortium Fellowship (Apple)	2024
Carnegie Mellon University Robotics Pathways Fellowship	2024

Undergraduate

Thurgood Marshall College Fund Apple Scholarship	2023
Crown Castle Opportunity Scholarship	2023
Howard University Capstone Scholarship	2020

COMMUNITY SERVICE

Panelist: Google CS Education Week (2023), PhD Virtual Information Session (2025)

Volunteer: NSBE Career Fair (2025), TAPIA Career Fair (2024), Provail (2026), PAMS Mentor (2025)

SELECTED MEDIA COVERAGE

<i>Grow with Google</i> , Watch what we do: Evolone Layne Grow with Google	2023
Google, Finding the Leader Inside: Evolone's Story as a GDSC Lead at Howard University	2023
<i>Glamour</i> , Evolone Layne Is Taking Over STEM, One App at a Time	2022
<i>BET</i> , Evolone Layne Is Taking Over STEM, One App at a Time	2022
<i>Hilltop</i> , Howard Student Named Glamour's College Women of the Year	2022